Cramer's Rule Practice

Use Cramer's rule to compute the solutions of the systems in Exercises 1–6.

1. $5x_1 + 7x_2 = 3$
$2x_1 + 4x_2 = 1$

2. $4x_1 + x_2 = 6$
$5x_1 + 2x_2 = 7$

3. $3x_1 - 2x_2 = 7$
$-5x_1 + 6x_2 = -5$

4. $-5x_1 + 3x_2 = 9$
$3x_1 - x_2 = -5$

5. $2x_1 + x_2 = 7$
$-3x_1 + x_3 = -8$
$x_2 + 2x_3 = -3$

6. $2x_1 + x_2 + x_3 = 4$
$-x_1 + 2x_3 = 2$
$3x_1 + x_2 + 3x_3 = -2$

In Exercises 7–10, determine the values of the parameter s for which the system has a unique solution, and describe the solution.

7. $6sx_1 + 4x_2 = 5$
$9x_1 + 2sx_2 = -2$

8. $3sx_1 - 5x_2 = 3$
$9x_1 + 5sx_2 = 2$

9. $sx_1 - 2sx_2 = -1$
$3x_1 + 6sx_2 = 4$

10. $2sx_1 + x_2 = 1$
$3sx_1 + 6sx_2 = 2$

Answers to odd-numbered problems

1. $\begin{bmatrix} 5/6 \\ -1/6 \end{bmatrix}$ **3.** $\begin{bmatrix} 4 \\ 5/2 \end{bmatrix}$ **5.** $\begin{bmatrix} 3/2 \\ 4 \\ -7/2 \end{bmatrix}$

7. $s \neq \pm\sqrt{3}$; $x_1 = \dfrac{5s+4}{6(s^2-3)}, x_2 = \dfrac{-4s-15}{4(s^2-3)}$

9. $s \neq 0, -1$; $x_1 = \dfrac{1}{3(s+1)}, x_2 = \dfrac{4s+3}{6s(s+1)}$